Data Complexity Measure

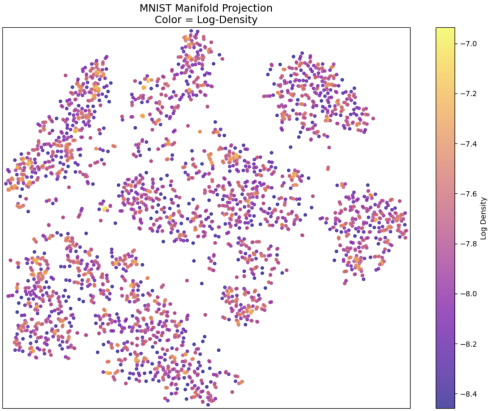
Problem

Existing measures scale linearly with n or rely on local estimates (LID), ignoring global geometry and duplication invariance.

Proposed Measure

$C(S) = \Phi(\{\mathbf{x}_i\}, \mathbf{s}_{\theta^*})$ via diffusion model score $\mathbf{s}_\theta \approx \nabla \log p_t$.
Properties:

- Duplication invariance: $C(S \cup \{\mathbf{x}\}) = C(S)$ if $\mathbf{x} \in S$.
- Monotonicity: $\mathcal{M}_1 \subseteq \mathcal{M}_2 \Rightarrow C(P_1) \leq C(P_2)$.
- Consistency: $C(S) \xrightarrow[n \to \infty]{} \mathcal{C}(P)$.



Projection of MNIST manifold with density estimation